

Multiple Person Tracking Comparison using Template Matching and Lucas-Kanade Optical Flow

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I. BACKGROUND

A. Introduction

This work compares two classical and complementary tracking approaches applied to pedestrian image sequences: a template-matching tracker based on patch similarity, and a Lucas–Kanade optical flow tracker that estimates local motion from image gradients. Template matching locates the most similar patch to a stored template using a pixel-wise similarity criterion; it is simple and interpretable but sensitive to appearance changes. The Lucas–Kanade method estimates per-pixel motion in a small neighborhood by solving a local least-squares problem; it is robust to small displacements and can exploit motion consistency but may fail when image gradients are weak or when the brightness constancy assumption is violated.

The experiments use a pedestrian dataset of short image sequences. Both trackers are initialized from the same bounding box in the first frame, and their predicted positions are compared frame-by-frame using geometric metrics and runtime measurements. This comparison highlights the trade-offs between accuracy, robustness, and computational cost for two representative classical approaches.

B. Objective

The objectives of this experiment are:

- Implement a straightforward template-matching tracker using a Sum-of-Squared-Differences (SSD) search within a local window and optional template updating.
- Implement a Lucas–Kanade tracker that estimates displacement from spatial and temporal intensity gradients and aggregates point-wise estimates to produce a single bounding-box translation.
- Quantitatively compare both methods using center error and Intersection-over-Union (IoU) relative to provided ground-truth bounding boxes, and compare wall-clock runtimes to illustrate computational trade-offs.

II. METHODOLOGY

A. Dataset

The experiments use the "Human Tracking & Object Detection Dataset" dataset available from Kaggle

(<https://www.kaggle.com/datasets/trainingdatapro/people-tracking>). This collection contains short image sequences of pedestrians captured in surveillance-like settings, together with annotation files that provide bounding boxes for persons in selected frames. In our workflow, frames are read in temporal order, converted to grayscale for intensity-based processing, and a single initial bounding box (from the first frame) is used to initialize both trackers. Ground-truth bounding boxes supplied by the dataset are used for per-frame quantitative evaluation.

B. Template Matching

The template-matching tracker locates the target by searching for the image patch most similar to a stored template. In this implementation the similarity measure is the Sum of Squared Differences (SSD):

$$\text{SSD}(x, y) = \sum_{i,j} (T(i, j) - I(x + i, y + j))^2,$$

where T denotes the template patch and I the current frame. The tracker restricts the search to a local window around the previous position to reduce computation and avoid spurious matches far from the expected location. After selecting the location with minimum SSD, the bounding-box center is recorded; optionally, the template can be refreshed with the patch at the newly detected location to adapt to gradual appearance changes.

Practical details:

- The template is extracted from the first frame using the initial bounding box.
- A search margin constrains candidate top-left coordinates to a neighborhood around the previous box.
- SSD is computed only for candidate patches that have the same size as the template.
- Template updating is optional but improves robustness to slow appearance changes.

C. Lucas–Kanade Optical Flow

The Lucas–Kanade tracker estimates motion by solving a local least-squares problem based on spatial and temporal intensity gradients. Under the brightness constancy assumption,

$$I_x u + I_y v + I_t = 0,$$

where I_x , I_y are spatial derivatives, I_t is the temporal derivative, and (u, v) is the flow vector. Collecting equations for all pixels in a window yields the normal equations

$$(A^\top A) \mathbf{d} = -A^\top \mathbf{b}, \quad \mathbf{d} = \begin{bmatrix} u \\ v \end{bmatrix},$$

with A formed from I_x and I_y and $\mathbf{b} = I_t$. The implementation computes central finite differences to approximate I_x and I_y , and per-pixel differences for I_t .

To obtain a single displacement for the tracked object, the tracker evaluates flow at a set of points sampled inside the initial ROI, computes the per-point (u, v) estimates, and aggregates them (median is used in the notebook) to produce a robust translation estimate for the bounding box center. The bounding box is then translated accordingly and clipped to remain inside image boundaries.

Practical details:

- Gradients are computed on the average of two consecutive frames to reduce noise.
- A small square window around each sample point is used to gather equations for the least-squares solve.
- A determinant threshold prevents unstable solutions when gradients provide insufficient information.
- The median of point-wise flows reduces the influence of outlier estimates from noisy or occluded points.

D. Evaluation Metrics

Both trackers produce a sequence of box centers per frame. Evaluation uses the center error (Euclidean distance between predicted and ground-truth centers) and Intersection-over-Union (IoU) computed between predicted boxes (using the initial box size) and ground-truth boxes. Wall-clock runtime is measured for each method to compare computational cost. The implementations favor clarity and reproducibility: they rely on simple, well-understood numerical steps and minimal external dependencies.

III. RESULT

The quantitative evaluation of the tracking experiments in Table I, conducted over a sequence of 41 frames, reveals a stark contrast between Template Matching (TM) and the Lucas-Kanade (LK) Optical Flow algorithm across all primary tracking and efficiency metrics. The trend is also visible in Fig. 1, which compares per-frame center error and per-frame IoU against ground truth. When examining spatial accuracy, the template matching approach demonstrates significantly greater precision and stability than its optical flow counterpart. Specifically, the template matching algorithm achieved a remarkably low mean center error of 6.43 px with a tight standard deviation of 2.62 px. In contrast, the Lucas-Kanade tracker yielded a mean center error of 13.05 px, more than double that of the template matching method, accompanied by a high standard deviation of 10.61 px. This substantial variance in the Lucas-Kanade error indicates that its tracking accuracy fluctuated heavily throughout the image sequence, suggesting a highly volatile tracking trajectory.

TABLE I
TRACKING METRICS SUMMARY: COMPARISON OF TEMPLATE MATCHING AND LUCAS-KANADE PERFORMANCE

Metric	TM	LK
Frames evaluated	41	41
Mean center error (px)	6.43	13.05
Std center error (px)	2.62	10.61
Max center error (px)	11.63	34.18
Mean IoU	0.779	0.650
Min IoU	0.617	0.298
Mean TM-LK distance (px)	15.66	
Runtime TM (s)	10.854	
Runtime LK (s)	0.475	
Runtime ratio TM/LK	22.844	

This volatility is further underscored by the maximum center error recorded during the sequence. While the template matching tracker remained tightly bound to the target with a peak error of only 11.63 px, the Lucas-Kanade tracker suffered from a severe drift phenomenon, reaching a maximum center error of 34.18 px. This spatial divergence between the two trajectories is captured by the mean distance between the template matching and Lucas-Kanade bounding box centers, which averaged 15.66 px across the sequence. The widening physical separation between the two tracking windows as the frames progressed points to a fundamental breakdown in the local motion estimation of the optical flow framework. The coordinate-level drift behavior can be observed clearly in Fig. 2, where horizontal and vertical position trajectories are compared against ground truth.

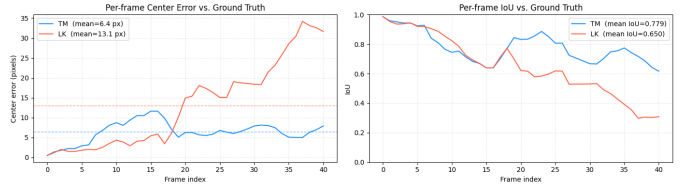


Fig. 1. Error-IoU Profile: per-frame comparison of Template Matching and Lucas-Kanade against ground truth for center error and IoU.

This degradation in spatial accuracy directly reflects on the structural alignment of the tracking windows, as quantified by the Intersection over Union (IoU) metrics. The template matching approach maintained a highly robust and reliable overlap with the target, yielding a mean IoU of 0.779. Even in its worst-performing frame, its minimum IoU never dropped below 0.617, ensuring that the bounding box consistently encapsulated the bulk of the target pedestrian. On the other hand, the Lucas-Kanade method produced a lower mean IoU of 0.650. More critically, its minimum IoU plummeted to a poor 0.298, a value that typically signifies a near-total tracking failure where the estimated bounding box retains only a marginal, peripheral overlap with the actual object.

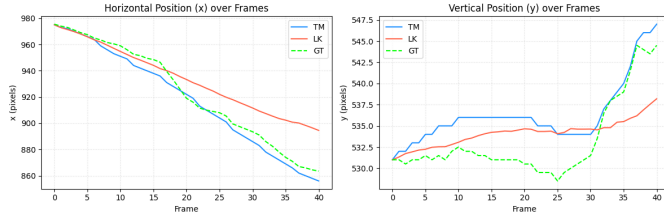


Fig. 2. Trajectory Profile: comparison of horizontal and vertical coordinate trajectories across frames.

However, shifting the focus to the temporal domain reveals a completely inverted performance profile, exposing the classic trade-off between spatial accuracy and computational efficiency in computer vision pipelines. The template matching algorithm required a total wall-clock runtime of 10.854 seconds to process the 41-frame sequence. Conversely, the Lucas-Kanade tracker completed the exact same sequence in a mere 0.475 seconds. This translates to an extraordinary runtime ratio of 22.844, proving that the Lucas-Kanade optical flow implementation executed nearly 23 times faster than the template matching framework. This runtime gap is summarized in the bar chart in Fig. 3.

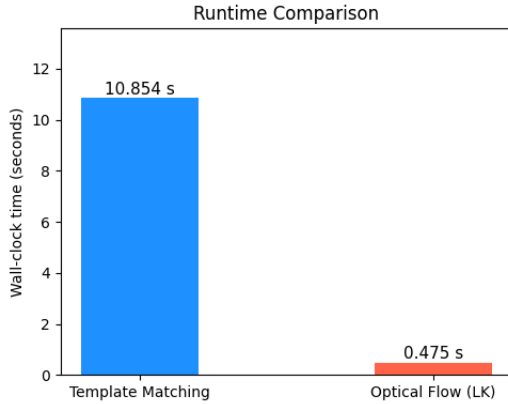


Fig. 3. Runtime Profile: execution-time comparison between Template Matching and Lucas-Kanade.

The underlying mechanics of both algorithms explain this dramatic divergence in speed and accuracy. Template matching, utilizing a Sum of Squared Differences (SSD) criterion, relies on an exhaustive spatial search. For every single frame, the algorithm must slide the target template pixel-by-pixel across a designated search region, computing dense intensity differences at every valid coordinate. While this brute-force approach guarantees that the globally optimal matching patch within that search space is selected, thereby preserving low center errors and high IoU scores, it imposes a heavy computational penalty that scales poorly and prevents real-time deployment on standard hardware.

In contrast, the Lucas-Kanade algorithm bypasses global image searching entirely by operating on local intensity gradients. Relying on the core assumptions of brightness constancy and spatial coherence, it sets up a sparse system of linear

equations for a localized neighborhood of keypoints, resolving the apparent velocity field using a closed-form least-squares formulation. This mathematical optimization is exceptionally lightweight, which explains its sub-second execution time. However, because it operates strictly on localized, differential pixel movements, it is inherently susceptible to tracking drift when applied to a dynamic pedestrian sequence. Pedestrians represent non-rigid bodies characterized by complex deformations, shifting limb postures, and subtle changes in scale or illumination. When these localized changes violate the brightness constancy assumption or exceed the neighborhood's motion boundaries, the Lucas-Kanade tracker loses its lock on individual feature points. As these errors compound over successive frames, the sparse points lag behind, causing the severe bounding box drift, elevated maximum center errors, and the sharp drop in minimum IoU observed in the final quantitative metrics.

As an illustrative example, we highlight a single target (Person ID=11). The initial trajectory and bounding boxes for ID=11 are shown in Fig. 4, while the final-frame tracker predictions compared to ground truth are shown in Fig. 5. These two example figures demonstrate how both methods behave on the same pedestrian instance and complement the aggregate metrics reported above.

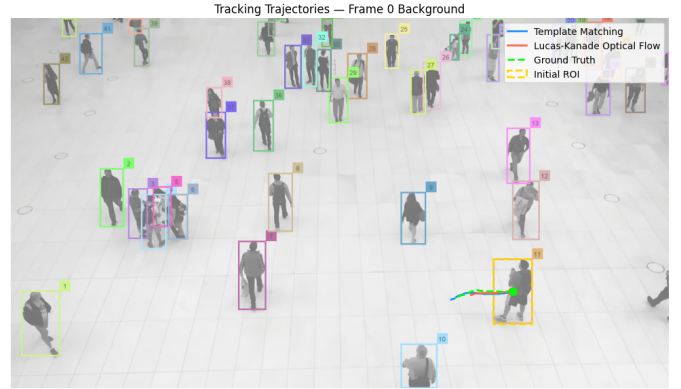


Fig. 4. Example (ID=11) — Tracking trajectories and bounding boxes on the frame 0 background.

IV. CONCLUSION

The evaluation of the tracking sequence reveals a classic computer vision trade-off where spatial accuracy and execution speed are completely inverted. Template Matching delivers far superior tracking precision, maintaining a highly stable bounding box over the pedestrian with a low average error of just 6.43 pixels and a robust 78% mean overlap. By relying on a thorough pixel-by-pixel search, it easily handles the structural changes of the target without losing its alignment. In contrast, the Lucas-Kanade framework struggles to adapt to the non-rigid deformations and shifting postures of a walking person. This causes its localized tracking to rapidly drift, leading to double the average positional error and a severe drop in frame-to-frame alignment, which occasionally falls to a marginal 30

Despite its tracking vulnerability, the Lucas-Kanade algorithm excels completely when it comes to computational efficiency. It processes the entire sequence in a mere 0.475 seconds, executing nearly twenty-three times faster than Template Matching, which bogs down the pipeline at over ten seconds. This incredible speed advantage stems from Lucas-Kanade's local gradient mathematics, bypassing the exhausting search areas required by the template-based approach. Ultimately, these results indicate that while Template Matching is the preferred choice for precise, offline tracking scenarios, Lucas-Kanade is far better suited for immediate, real-time deployments where speed is critical and minor spatial drift can be compensated for by secondary correction filters.



Fig. 5. Example (ID=11) — Final-frame tracker predictions vs. ground truth for person ID=11.